



Course: Chemical Evolution of Galaxies

Do we expect the Hubble sequence to be present at all redshift?

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Date Last Edited: April 22, 2024

The Hubble morphological classification system describes the shapes of galaxies in the local universe (Hubble 1926). This system not only depicts the visual appearance of these galaxies but has also shown that the corresponding morphological type strongly correlates with properties such as the star formation rate, age, number of merger events, etc. However, despite the success of the Hubble diagram in describing galaxies at very high redshifts $z \geq 7$ (Ferreira et al. 2023), the number of irregular galaxies increases rapidly with redshift, thus requiring alternative descriptions (Tohill, Bamford, and C. Conselice 2023).

Elliptical galaxies tend to be redder in color and are systems that do not form stars, while galaxies with disk morphologies tend to be bluer and have a higher star formation rate (e.g., Sandage 1986, Strateva et al. 2001, Kauffmann et al. 2003, Christopher J. Conselice 2006, Nair and Abraham 2010). Regarding galactic structure, elliptical galaxies exhibit higher concentrations, lower asymmetries, and larger Sérsic indices than disk galaxies at low redshifts (e.g., Christopher J. Conselice 2003, Christopher J. Conselice, Blackburne, and Papovich 2005, Scarlata et al. 2007, Nair and Abraham 2010).

At redshifts $z < 1$, most bright ($M_B < -20$) and massive ($M_* > 10^{10}$) galaxies are primarily ellipticals and spirals, persisting until approximately $z \sim 1$ with minimal variations. The co-moving density of these galaxies remains largely constant up to $z \sim 1$, as observed by Brinchmann and Ellis 2000 and Christopher J. Conselice, Blackburne, and Papovich 2005. However, other features of normal galaxies show more significant changes from $z \sim 1$ to $z \sim 0$, including the co-moving B-band luminosity densities and stellar mass densities. The decline in B-band luminosity density for galaxies of known morphology at $z < 1$ suggests decreased luminosity densities overall, attributed to recent star formation evidenced by the increase in stellar mass density. In Hubble Deep Field-North (HDF-N), the stellar mass density for ellipticals is half its present value at $z \sim 1$, indicating a possible overabundance of ellipticals at that redshift. Studies on galaxies in the 'red sequence' support this evolution, with stellar mass doubling from $z = 1$ to $z = 0$, indicating ongoing star formation primarily in lower mass and luminosity systems. In summary, while higher mass systems may have formed by $z \sim 1$ or earlier, star formation continues to influence the evolution of lower mass and luminosity systems up to $z < 1$.

The redshift interval $1 < z < 2$ plays a crucial role in comprehending the final emergence and formation of the Hubble sequence and the origin of the galaxy structure-redshift relationship. Galaxies at higher redshifts exhibit notably peculiar and clumpy morphologies, as evidenced in studies such as Noguchi 1998 and Christopher J. Conselice, Blackburne, and Papovich 2005. This phenomenon arises from various factors, some of which are well-understood, such as the cosmic star formation history. It's established that the universe experienced its peak star formation rate around $z \sim 2$, implying that galaxies would have possessed more star-forming regions, resulting in clumpier morphologies. Additionally, higher merger rates during these earlier epochs, associated with AGN activity and stellar mass assembly, led to galaxies displaying disturbed morphologies with tidal disruptions and multiple cores, among other features. However, what remains elusive is the mechanism driving the evolution of these galaxies into the ones observed today. This evolutionary process can be explored through the study of morphological changes with redshift (Tohill, Bamford, and C. Conselice 2023).

Finally for higher redshifts it becomes less clear. In Mortlock et al. 2013, it was found that beyond a redshift of $z > 2$, the universe is primarily characterized by peculiar galaxies, although a significant population of spheroids persists, indicating the early presence of mechanisms governing the formation of the Hubble sequence. Not until $z_{\text{trans}} = 1.86 \pm 0.62$ (transition redshift where the combined fraction of spheroidal and disc galaxies is equal to that of the peculiar population) does the universe begin to exhibit the morphological diversity observed locally. The scarcity of galaxies displaying visual disk morphologies at $z > 2$ suggests that the Hubble tuning fork model may not accurately apply to galaxies at these high redshifts. One possible explanation is that these disks have already formed but are undergoing disruption due to various processes that influence galaxy evolution at high redshifts. Simulation studies (e.g., Dekel, Sari, and Ceverino 2009, Ceverino, Dekel, and Bournaud 2010) indicate that rotating disks can experience violent instabilities, leading to irregular, clumpy morphologies.

Research on the prevalence and dominance of Hubble type galaxies at high redshifts, particularly beyond $z > 2$, presents conflicting findings. Some studies (e.g., Dickinson 2000, Papovich et al. 2005, Cameron et al.

2011) suggest a scarcity of such galaxies, indicating significant evolutionary processes. Conversely, other studies (e.g., Driver et al. 1998, Christopher J. Conselice, Blackburne, and Papovich 2005, Szomoru et al. 2011, C. J. Conselice et al. 2011, Buitrago et al. 2011) identify a dominant peculiar population alongside Hubble type galaxies. Discrepancies may arise from factors like sample selection and image quality. Resolving these discrepancies requires further investigation with a comprehensive sample and detailed imaging. Such efforts will help address biases and provide clearer insights into the evolution of galaxies at high redshifts.

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